Theodor Spiro

Computational biophysics — cell states in cancer and stress

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RESEARCH INTERESTS

How a cell integrates many signals into a single decision — to divide, to change identity, or to die — and what changes in that process in cancer. The main result so far is a reference map of normal human cell states, built without any cancer data, on which malignant cells from 25 cancer types converge on one restricted region; within a patient, malignant epithelium occupies that state far more often than the patient's own normal tissue. The method is computational analysis of single-cell data, with systematic testing of whether a result reflects biology or a technical artifact. The next step requires experiments: measuring directly how cells move between these states under stress.

EDUCATION

Lomonosov Moscow State University, Faculty of Physics, Moscow

M.Sc. program in Physics (Department of Biophysics) · 2021 – 2022
One semester completed: graduate courses in physics of biopolymers, magnetic resonance spectroscopy in biology and medicine, computer modelling in biology, and physicochemical kinetics. Studies interrupted on leaving Russia in 2022.

B.Sc. in Physics (Department of Biophysics) · 2014 – 2021
Thesis, graded excellent: *Effects of Jerusalem artichoke polysaccharide on cell state in a cellular automaton model.* Supervisor: Prof. Leonid Yakovenko. Completed over an extended period alongside teaching work (see Teaching) and family responsibilities.

RESEARCH EXPERIENCE

Independent researcher · 2022 – present
Affiliated with Vaika Inc. (nonprofit aging-research organization), East Aurora, NY · 2024 – present
Independent computational research program on cell states in cancer and aging. Built a reference coordinate system of normal human cell states and showed that malignant cells from 25 cancer types converge on one region of it, and that within a patient malignant epithelium occupies that state far more often than the patient's own normal tissue. Developed open-source software for this analysis and for separating biological signal from technical artifacts in single-cell data. Manuscript preparation advised by Dr. Andrei Gudkov (Roswell Park Comprehensive Cancer Center).

Undergraduate research, Department of Biophysics, Moscow State University · 2018 – 2021
Supervisor: Prof. Leonid Yakovenko. Stochastic cellular-automaton model of how a cell chooses between survival and apoptosis after receptor stimulation (Dectin-1, TLR6, TNFR1), including the NF-κB and c-FLIP feedback that controls caspase-8 activation. The model predicted three regimes — apoptosis, survival, and unstable oscillation between them — depending on the strength of that feedback.

PREPRINTS AND MANUSCRIPTS

All single-author; none peer reviewed.

Preprints

1. **Diverse cancers converge on one region of normal cell-state space in a cancer-blind coordinate system.** Zenodo, 2026. doi:10.5281/zenodo.20542129
   A map of 154 normal human cell types built without cancer data; malignant cells from 25 cancer types fall into one restricted region. Within patients, tumor epithelium occupies that state far more often than the same patient's normal epithelium (colorectal: 43 of 46 patients), in every tissue tested except prostate.

2. **Waveform asymmetry as a biomarker of neural aging.** Zenodo, 2026. doi:10.5281/zenodo.19912202
   The shape of brain-wave cycles changes with age in two independent cohorts of more than 800 adults, more strongly than the classical slowing of the alpha rhythm.
3. **Spectral exponents of the twelve-lead ECG reveal the anatomy of cardiac conduction disorders.** Zenodo, 2026. doi:10.5281/zenodo.19945065
   In 412,000 electrocardiograms from three continents, one measure of the signal's frequency structure identifies the type of conduction block; it reflects cardiac anatomy rather than aging.
4. **Transcriptomic noise accumulates within tissue identity across human aging.** Zenodo, 2026. doi:10.5281/zenodo.19944444
   With age, gene activity in human tissues becomes noisier while each tissue keeps its identity; caloric restriction reduces the noise rather than restoring structure.
5. **Universal statistical signatures of evolution in artificial intelligence architectures.** arXiv:2604.10571, 2026.
   Modifications of neural-network architectures show a distribution of effects similar to that of mutations in living organisms.

Manuscripts in preparation — cancer signaling

1. **Temporal architecture of signaling oscillations predicts cancer gene function.** Across 14 signaling pathways, genes that switch a signal on are enriched for oncogenes, and genes that switch it off for tumor suppressors.
2. **Negative feedback loop architecture predicts cancer vulnerability.** Genes in negative feedback loops of signaling pathways are strongly enriched for known cancer genes.

Other work. Further manuscripts and pilot studies — proteostasis in memory consolidation, neural-network interpretability, forecasting, creativity measurement — and two negative results reported in full are listed at mool32.github.io/publications (https://mool32.github.io/publications/).

SOFTWARE

perceptome (2026). Open-source Python package that places any cell on the reference map of normal cell states and runs the within-patient convergence assay, with built-in checks against common artifacts. doi:10.5281/zenodo.20113468

metric-autopsy (2026). Open-source tool that tests, step by step, whether a measurement taken from single-cell data reflects biology or a technical artifact; built from three of the author's own documented analysis errors. doi:10.5281/zenodo.21195679

GRANTS AND HONORS

Emergent Ventures grant, Mercatus Center at George Mason University · 2026

PRESENTATIONS AND MEETINGS

Lomonosov Scientific Conference, Moscow · 2021 · talk on the cellular-automaton model of the survival–apoptosis decision

Emergent Ventures unconference, Copenhagen · 2026 · participant

TEACHING

Mathematics and physics to students aged 11–25, including IB and A-level examination and olympiad preparation.

- **Le Sallay Academy**, international program for gifted students, including the IB curriculum: curriculum design and teaching · 2025 – 2026
- **BrainyBara** (Israel): co-founder and lead teacher · 2022 – present
- **Moscow State University**: instructor, preparatory courses and olympiad mathematics · 2018 – 2020
- **Foxford Online School** (Moscow): teacher · 2015 – 2018

Evidence-Based Teaching Practices, Derek Bok Center for Teaching and Learning, Harvard University — online course with certificate · 2022 – 2024

SKILLS

Computational. Python; statistics; single-cell RNA sequencing analysis; mathematical modelling (cellular automata, dynamical systems); open-source research software with automated testing.

Research practice. Predictions registered before analysis; negative results reported in full; AI tools used for coding and analysis, with independent validation of every result.

LANGUAGES

Russian (native) · English (fluent) · Hebrew (basic)

REFERENCES

Available on request.